# Software Requirements Specification (SRS)

## 1. Introduction

### 1.1 Purpose

The purpose of the STEAMROLLER system is to help people who stutter communicate more clearly by converting their speech into fluent and understandable speech.

The system is designed to:

- Convert stuttered speech into fluent speech
- Remove repetitions and errors
- Keep the original meaning
- Improve communication in daily use

### 1.2 Problem Description

**a) Purpose of the SRS**

The purpose of this SRS is to describe the requirements and working of the STEAMROLLER system, which helps people who stutter by converting their speech into fluent and clear speech.

**b) Intended Audience**

This document is intended for:

- Software developers to design and implement the system
- Testers to verify system functionality and performance
- Project supervisors and evaluators to understand the system design and objectives
- Future developers or researchers who may extend or improve the system

### 1.3 Scope

**a) Software Product Name**

The software product is **STEAMROLLER**, a system that converts stuttered speech into fluent speech.

**b) What the System Will and Will Not Do**

**The system will:**

- Take speech as input
- Convert it into text
- Fix stuttering errors
- Generate clear speech

**The system will not:**

- Cure stuttering
- Work without required tools or internet

**c) Application, Benefits, and Goals**

The system is used in communication applications like meetings and interviews.

**Goals:**

- Improve communication
- Increase accuracy
- Provide fluent speech output

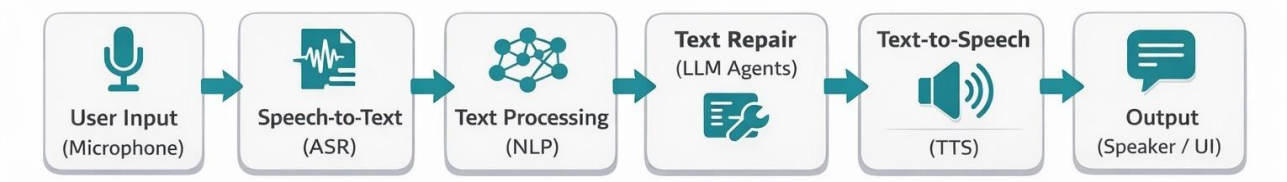
# 2. Overall description

This section describes the overall structure and basic requirements of the STEAMROLLER system.

## 2.1 Product perspective

The STEAMROLLER system is a speech processing system that works as a pipeline. It takes stuttered speech as input, processes it using AI modules, and produces fluent speech as output.

### 2.1.1 Block Diagram




The system consists of the following main components:

- Speech Input Module (Microphone/User Input)
- Speech-to-Text Module (ASR)
- Text Processing Module (NLP)
- Multi-Agent Text Repair Module
- Text-to-Speech Module (TTS)
- User Interface

2.1.2. Hardware Requirements

2.1.3. Software Requirements

### 2.1.2 Hardware Requirements

- Computer/Laptop
- Minimum 8GB RAM (16GB recommended)
- Processor: Intel i5 / Ryzen 5 or higher
- Microphone for audio input
- GPU (optional but recommended)

### 2.1.3 Software Requirements

- Operating System: Windows/Linux/Mac
- Programming Language: Python
- Tools: VS Code / PyCharm

- Libraries: PyTorch, Transformers
- Models/APIs:
    - ASR (Whisper)
    - LLM (for text repair)
    - TTS (StyleTTS or similar)

### 2.2 Product functions

**The STEAMROLLER system performs the following main functions:**

- Accepts stuttered speech as input
- Converts speech into text using ASR
- Detects and removes stuttering errors
- Repairs text while keeping original meaning
- Converts corrected text into fluent speech
- Provides near real-time output

## 3. Specific requirements

This section describes the detailed requirements of the STEAMROLLER system, including inputs, outputs, and system behaviour.

### 3.1 External interfaces

**Inputs:**

- Stuttered speech (audio input from microphone or file)

**Outputs:**

- Fluent speech (audio output with preserved voice)

**User Interface:**

- Simple web interface
- Options to record or upload audio
- Display of original and corrected audio

### 3.2 Functional Requirements

The system shall:

- Accept speech input from the user
- Convert speech into text using ASR
- Detect stuttering such as repetitions and pauses
- Remove unnecessary repetitions and errors
- Correct grammar and sentence structure
- Preserve the original meaning of speech
- Convert corrected text into fluent speech
- Provide output in near real-time

## 3.3 Performance requirements

**Static Requirements:**

- The system shall support multiple users (at least 5–10 users)
- The system shall handle audio input up to a few minutes

**Dynamic Requirements:**

- The system should respond within 4-5 seconds
- The system should process speech continuously
- The system should maintain good accuracy in output

## 3.4 Design constraints

- The system depends on AI models (ASR, LLM, TTS)
- Requires internet connection for APIs
- Performance depends on hardware (CPU/GPU)
- Limited availability of stuttering datasets

## 3.5 Logical database requirements

**Data Stored:**

- Audio input (stuttered speech)
- Intermediate text (temporary processing only)
- Final output audio (fluent speech with preserved voice)

**Details:**

- Audio data is used for processing and may be stored temporarily
- Text is generated only for internal processing (not final output)
- Final output is fluent audio that preserves the speaker's voice characteristics
- No permanent storage of user data to ensure privacy

**Data Flow Structure:**

User → Audio Input → Text (Processing) → Repaired Audio Output

## 3.6 Software system attributes

The quality features of the STEAMROLLER system that ensure it works properly and efficiently.

### 3.6.1 Reliability

- The system should produce accurate and consistent results
- It should correctly handle different types of stuttering
- It should not fail during normal operation
- Error handling should be available for invalid inputs

### 3.6.2 Availability

- The system should be available most of the time (high uptime)
- It should quickly recover from failures
- The system should support continuous usage during operation

### 3.6.3 Security

- User audio data should be protected
- No unauthorized access to the system
- Secure communication with APIs and models
- No permanent storage of sensitive user data

### 3.6.4 Maintainability

- The system should be easy to update and improve
- Code should be modular and well-structured
- Bugs and errors should be easy to fix
- **New features can be added easily**

### 3.6.5 Portability

- The system should run on different platforms (Windows, Linux)
- It should be deployable on web and cloud environments
- Can be extended to mobile or edge devices

# 4. Software Design Document

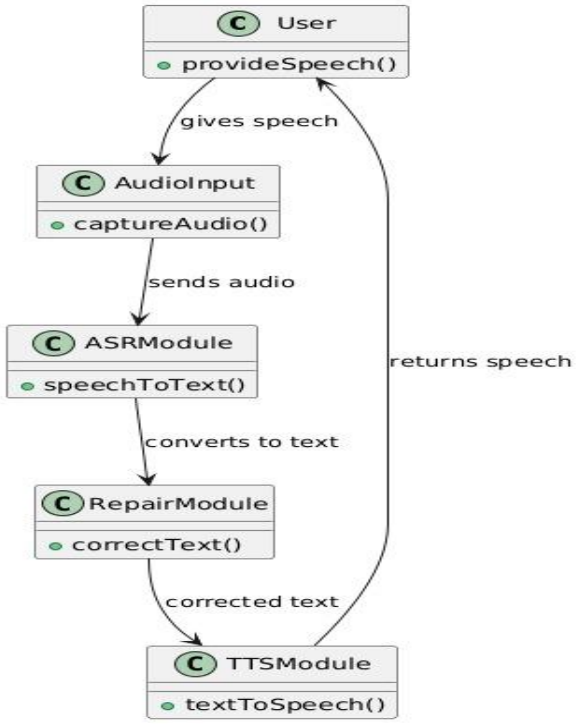
This section describes the design of the STEAMROLLER system using different diagrams to explain structure, data flow, and behavior.

## 4.1. Structural Design

### a. Class Diagrams

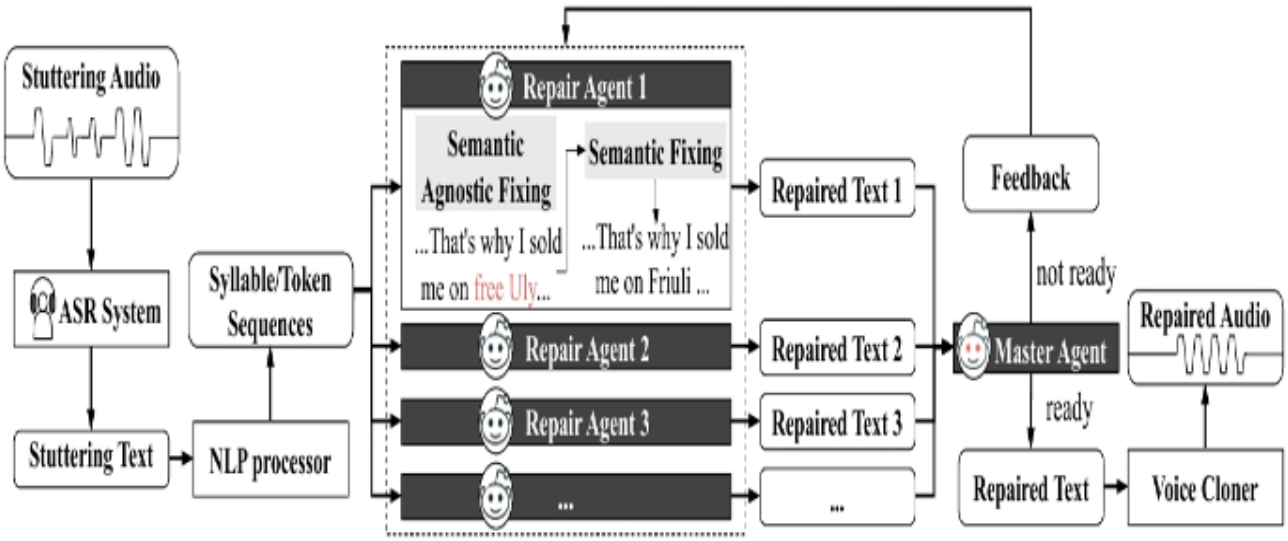
It shows the main classes of the system such as:

- User
- Audio Input
- ASR Module
- Repair Module (Agents)
- TTS Module



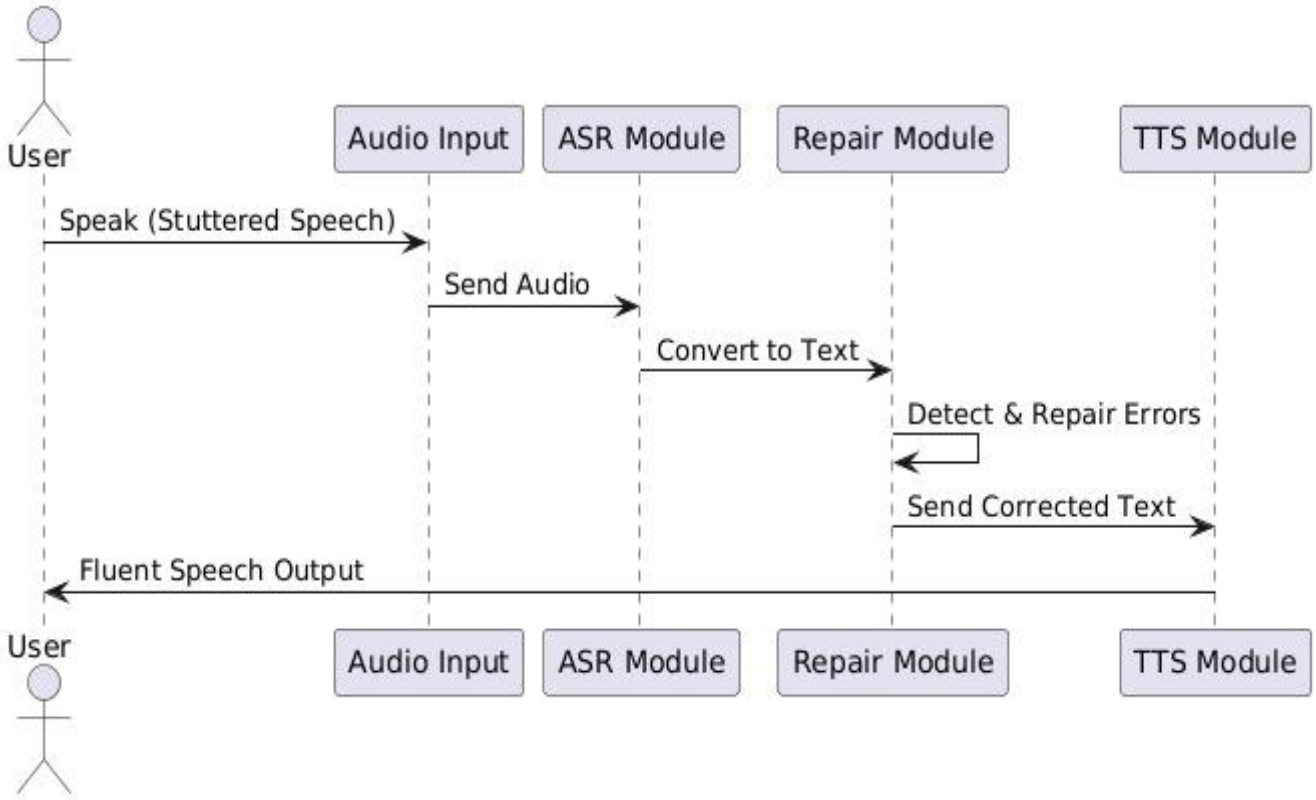
### 4.2. Data Design

**a. Data-flow Diagram :** It shows how data flows in the system.

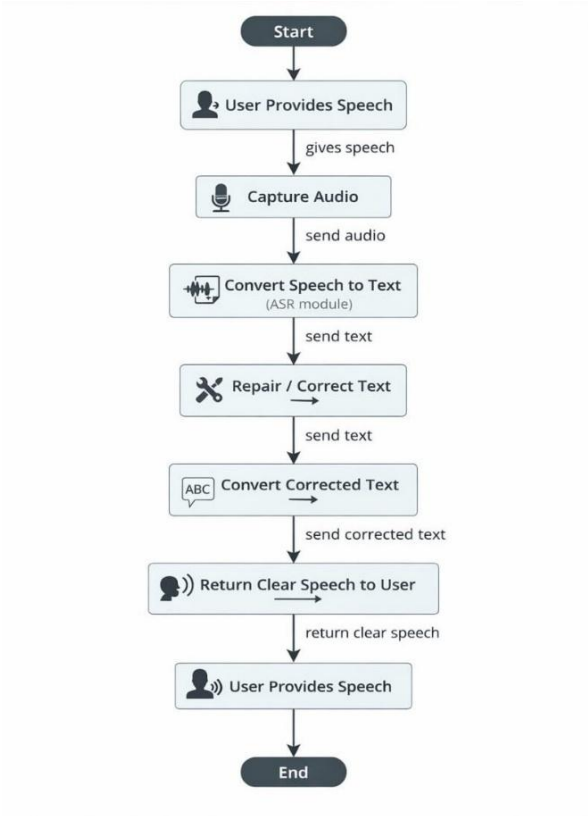



### 4.3. Behavioural Design

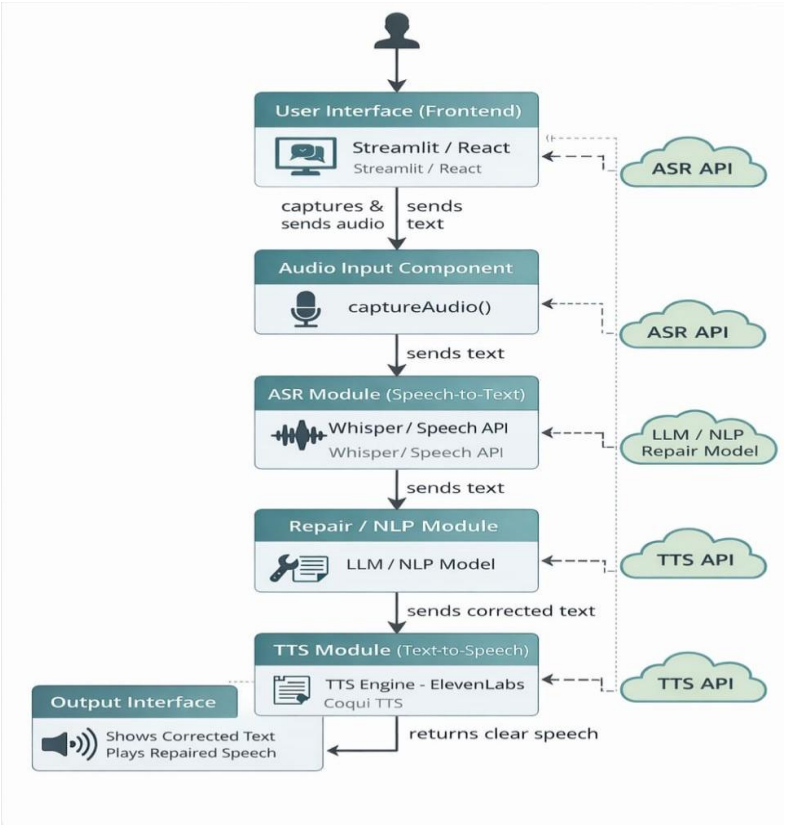
**a. Sequence Diagram :** It shows step-by-step interaction.



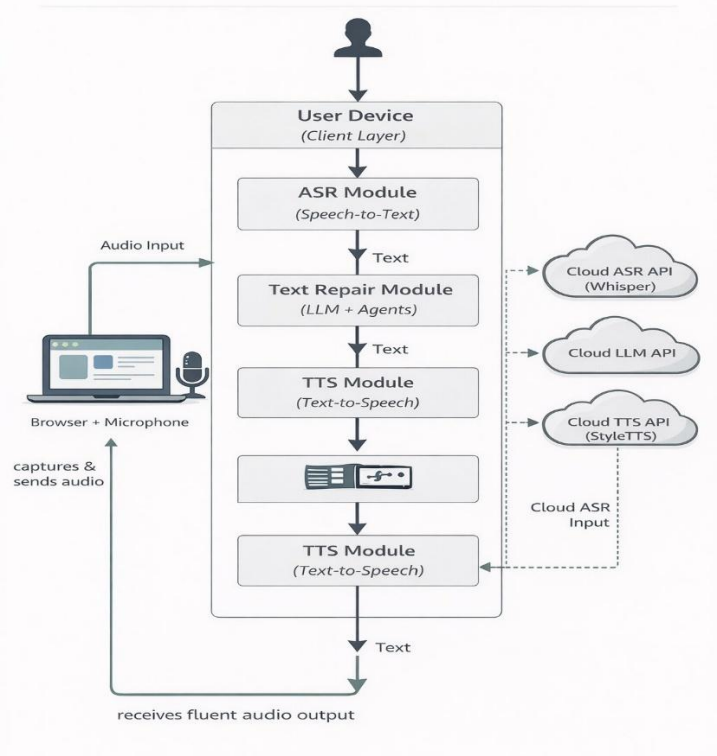
**b. Activity Diagram**




**4.4.a Implementation Design : Component Diagram –** It show all the components.



**4.4.b Environment Design : Deployment Diagram -** It shows where system runs.




# 5. References

**References List:**


1. **STEAMROLLER: A Multi-Agent System for Inclusive Automatic Speech Recognition for People Who Stutter**
    - Type: Research Paper
    - Date: 2026
    - Source: arXiv
2. **Robust Speech Recognition via Large-Scale Weak Supervision**
    - Authors: Alec Radford et al.
    - Conference: ICML
    - Year: 2023
3. **StyleTTS 2: Towards Human-Level Text-to-Speech through Style Diffusion**
    - Conference: NeurIPS
    - Year: 2023
4. **FluencyBank: A Resource for Fluency Research**
    - Journal: Journal of Fluency Disorders
    - Year: 2018
5. Project Documents
    - Project Synopsis (STEAMROLLER)
    - Project Presentation (PPT)


**Source Access:**

- Research papers are available on platforms like arXiv and IEEE Xplore
- Project documents are provided internally by the project team

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